9 Ko’phadlar ustida amallar

1-to’plam. *Qo’shish/ayirish – ayirish ishorasi qopqoniga e’tibor bering.*

1. $(3x^2 + 2x - 5) + (x^2 - 4x + 7) =$

2. $(5x^3 - 2x + 1) - (2x^3 + 3x - 4) =$

3. $(-2x^2 + 6x - 3) + (4x^2 - 6x + 8) =$

4. $(6x^2 - 3x) - (2x^2 - 3x + 5) =$

5. $(x^3 + 2x^2 - x) + (-x^3 + 5x - 3) =$

6. $(4x^2 + 7) - (x^2 - 2x + 7) =$

2-to’plam. *Binomlarni ko’paytirish (FOIL usuli).*

1. $(x + 3)(x + 5) =$

2. $(x - 2)(x + 7) =$

3. $(2x + 1)(x - 4) =$

4. $(x - 6)(x - 6) =$

5. $(3x + 2)(3x - 2) =$

6. $(x + 4)(2x^2 - x + 3) =$

3-to’plam. *Qiyinroq ko’paytirish – kvadratlar, binom×trinom, uch ko’paytuvchi.*

1. $(x+2)^2 =$

2. $(2x-1)^2 =$

3. $(x^2+3x-2)(x+4) =$

4. $(x-1)(x^2+x+1) =$

5. $(3x+1)(x^2-2x+5) =$

6. $(x+2)(x-2)(x+1) =$

4-to’plam. *Bo’lish.*

1. $(6x^3-9x^2+3x) \div 3x =$

2. $(8x^2+12x) \div 4x =$

3. $(10x^4-5x^3+15x^2) \div 5x^2 =$

4. $(x^2+5x+6) \div (x+2) =$

5. $(x^2-9) \div (x-3) =$

6. $(2x^2+7x+3) \div (2x+1) =$

5-to’plam. *Aralash – bir masalada bir nechta amal, haqiqiy test zichligi.*

1. $(x+3)(x-2)-(x^2-5)=$

2. $(2x-1)(x+4)+(x^2+3x)=$

3. $(x+5)^2-(x^2+10)=$

4. $(3x^2-2x+1)-(x+2)(x-1)=$

5. $2(x+1)(x-1)+(x^2-3)=$

6. $(x-4)^2+(8x-10)=$

6-to’plam. *Xatoni toping – buzilgan qadamni ayting, to’g’ri javobni bering.*

1. $(x^2+3x)-(x^2-2x+5)$ ifodasida ko’rsatilgan yechimda xato bor:

   “$x^2+3x-x^2-2x+5=x+5$.”

   Xato va to’g’ri javob: .....

2. $(x+3)(x-4)$ ifodasida ko’rsatilgan yechimda xato bor:

   “$x\times x+3\times(-4)=x^2-12$.”

   Xato va to’g’ri javob: .....

3. $3x\times(-5x)$ ifodasida ko’rsatilgan yechimda xato bor:

   “$=+15x^2$.”

   Xato va to’g’ri javob: .....

4. x^2+x^2 ifodasida ko’rsatilgan yechimda xato bor:

   “$=x^4$.”

   Xato va to’g’ri javob: .....

5. $(8x^2+4x)\div 4x$ ifodasida ko’rsatilgan yechimda xato bor:

   “$8x^2\div 4x=2x$, *demak javob* $2x+4x=6x$.”

   Xato va to’g’ri javob: .....

7-to’plam. *Bonus / qiyin – har biriga taxminan 90 soniya.*

1 $(x+1)(x+2)(x+3) =$

2 $(x^2+2x+1) \div (x+1) =$

3 $(3x-2)(2x+5) - (x-1)(x+6) =$

4 $(x+4)^2 - (x-4)^2 =$

5 $(2x^2-3x+1) \div (2x-1) =$

Javoblar

To’plam	#	Javob	#	Javob
1-to’plam	1	$4x^2 - 2x + 2$	2	$3x^3 - 5x + 5$
	3	$2x^2 + 5$	4	$4x^2 - 5$
	5	$2x^2 + 4x - 3$	6	$3x^2 + 2x$
2-to’plam	1	$x^2 + 8x + 15$	2	$x^2 + 5x - 14$
	3	$2x^2 - 7x - 4$	4	$x^2 - 12x + 36$
	5	$9x^2 - 4$	6	$2x^3 + 7x^2 - x + 12$
3-to’plam	1	$x^2 + 4x + 4$	2	$4x^2 - 4x + 1$
	3	$x^3 + 7x^2 + 10x - 8$	4	$x^3 - 1$
	5	$3x^3 - 5x^2 + 13x + 5$	6	$x^3 + x^2 - 4x - 4$
4-to’plam	1	$2x^2 - 3x + 1$	2	$2x + 3$
	3	$2x^2 - x + 3$	4	$x + 3$
	5	$x + 3$	6	$x + 3$
5-to’plam	1	$x - 1$	2	$3x^2 + 10x - 4$
	3	$10x + 15$	4	$2x^2 - 3x + 3$
	5	$3x^2 - 5$	6	$x^2 + 6$
7-to’plam	1	$x^3 + 6x^2 + 11x + 6$	2	$x + 1$
	3	$5x^2 + 6x - 4$	4	$16x$
	5	$x - 1$	–	–
6-to’plam	1. Faqat birinchi had ishorasi o’zgartirildi, ikkinchi qavs ichidagi barcha hadlar ishorasini almashtirish kerak. To’g’ri: $5x - 5$.			
	2. Faqat Birinchi va Oxirgi hadlar ko’paytirildi (FOIL to’liq emas) – Tashqi va Ichki hadlar tushib qolgan. To’g’ri: $x^2 - x - 12$.			
	3. Manfiy×manfiy musbat berishi kerak, bu yerda musbat×manfiy bo’lgani uchun natija manfiy. To’g’ri: $-15x^2$.			
	4. Darajalar ko’paytirishdagidek qo’shildi; bu o’xshash hadlarni qo’shish, daraja o’zgarmaydi. To’g’ri: $2x^2$.			
	5. Faqat birinchi had $4x$ ga bo’lindi, ikkinchi had ham bo’linishi kerak. To’g’ri: $2x + 1$.			

Video yechimlar va qo’shimcha testlar uchun quyidagi QR kodlarni skanerlang:

QR kod

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